
Automated Text Simplification for Easy Read Generation

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Abstract

Easy Read is an accessibility format designed to make written information more understandable for people with intellectual disabilities, but producing high-quality Easy Read documents requires trained specialists and does not scale easily to large collections of text. Recent large language models (LLMs) offer a promising approach to automated simplification, yet naive prompting can drop important concepts, weaken semantic faithfulness, or fail to follow the specific syntactic and structural constraints of Easy Read writing. This paper presents a graph-guided LLM pipeline for Easy Read text simplification. We represent standard and Easy Read texts using Abstract Meaning Representation (AMR) graphs, train a Graph Neural Network (GNN) classifier to learn structural differences between the two forms, and convert graph-derived features into human-readable guidance for an LLM. A separate knowledge graph is used to rank central concepts and encourage concept preservation during generation. We compare this graph-guided approach against a baseline LLM prompt using SARI, BLEU, Flesch-Kincaid Grade Level, concept preservation, and average sentence length. Results show that graph-guided prompting improves SARI, BLEU, and concept preservation relative to the baseline while maintaining comparable readability levels. These findings suggest that graph-based semantic structure can help guide LLMs toward more faithful and accessible Easy Read simplifications.

1 Introduction

Access to written information is a fundamental requirement for civic participation, decision-making, and everyday life, yet individuals with intellectual disabilities often face difficulties reading complicated texts, creating persistent obstacles in their daily lives. Easy Read is widely recognized as a solution to improve the accessibility of complex documents and requires trained specialists to rewrite them according to established guidelines: short sentences, plain vocabulary, active voice, and concrete phrasing. While this process ensures the production of reliable, simplified documents, it is also expensive and slow, and does not scale well to the massive volume of documents that should be made more accessible.

Automated text simplification offers a promising alternative, and recent advances in Large Language Models (LLMs) have demonstrated strong capabilities in rewriting text at lower complexity levels. However, naive LLM-based simplification often treats a document as a flat sequence of sentences, ignoring its underlying conceptual structure. This can lead to outputs that drop key concepts, weaken semantic faithfulness, and lose important information from the original document. Furthermore, Easy Read texts obey very specific grammatical, structural, and syntactic constraints that are often overlooked by LLMs. This can produce simplified texts that are not well-tailored to the needs of users with intellectual disabilities.

This project explores graph-based machine learning methods to enhance the quality and accuracy of automated document simplification compared to naive LLM prompting, with a focus on the needs of users with intellectual disabilities. Easy Read production can also involve broader design choices, including page layout, spacing, visual supports, and image selection; these are important accessibility concerns but fall outside the scope of this project, which focuses on textual simplification, including sentence structure, grammar, and concept preservation.

Using Abstract Meaning Representation (AMR) graphs to represent the semantic relationships in both Easy Read and non-Easy Read texts (which we will refer to as standard text in this paper for simplicity), this project explores how effective Graph Neural Networks (GNNs) are at classifying AMRs as Easy Read or standard. Drawing from a parallel corpus of Easy Read and standard text, with their corresponding AMRs, we train a GNN to learn structural patterns that distinguish the two types of AMR graphs. These graph-derived features are then used to guide an LLM toward more faithful and accessible Easy Read simplifications. We evaluate the effectiveness of our graph-guided LLM text simplification output by comparing it to a naive LLM text simplification baseline using language evaluation metrics such as SARI, BLEU, Flesch-Kincaid Grade Level estimates of reading difficulty, and concept preservation. Results show that graph-guided prompting improves SARI, BLEU, and concept preservation compared to the baseline, while maintaining similar readability levels.

2 Background and Related Work

This project builds on prior work in Easy Read text generation, graph neural networks, AMR-based language generation, and graph-enhanced prompting for large language models. Together, these areas motivate our use of AMR-derived document graphs and GNN-based structural guidance for Easy Read simplification.

Easy Read text is designed to make written information more accessible to people with reading difficulties, including people with intellectual disabilities. Unlike general text simplification, Easy Read requires not only simpler vocabulary and grammar, but also structural constraints such as short sentences, natural line breaks, and segmentation at points where a reader would naturally pause. Calleja et al. [Calleja et al., 2024] show that automated Easy Read segmentation is viable but still falls short of expert-driven human segmentation, motivating our focus on sentence structure and syntactic simplification as a crucial component of Easy Read generation.

Our project uses graph neural networks to learn from document-level AMR graphs. Kipf and Welling [Kipf and Welling, 2017] introduce Graph Convolutional Networks (GCNs), which learn from both node features and graph structure through message passing. This makes GCNs suitable for AMR-based representations, where nodes correspond to semantic concepts and edges encode relations between them. Song et al. [Song et al., 2018] show that directly modeling AMR graph structure can improve AMR-to-text generation compared to relying only on linearized AMR strings, supporting our use of AMR graphs as structured semantic inputs for downstream simplification.

This is further supported by recent work revisiting the role of semantic graphs in the age of LLMs. Yao et al. [Yao et al., 2024] show that AMR-based semantic representations can still provide useful structure for syntactic simplification, particularly when used to guide LLM prompting through explicit graph-based reasoning. Our work follows this direction, but instead of prompting directly with AMR graph reasoning steps alone, we train a GNN over AMR-derived document graphs and convert graph-derived features into simplification guidance.

3 Data Collection and Preprocessing

The dataset used in this project comes from the Easy Read segmentation corpus associated with Calleja et al. [Calleja et al., 2024]. The dataset was provided courtesy of Thierry Etchegoyhen, one of the paper’s authors. At the time of this project, the dataset is not publicly released and must remain private. Although the full corpus includes Basque, English, and Spanish text files, we use the English portion for this project.

The raw data consists of plain text files containing Easy Read sentences with explicit `_seg_` markers, which indicate where a sentence should be split into Easy Read segments. Since our system requires

paired standard and Easy Read text, we treat the sentence with segmentation markers removed as `normal_text`, and the segmented version as the Easy Read reference.

Preprocessing was performed in three stages. First, `segmentation.py` converted the raw segmented text into JSONL records containing an integer `id`, the unsegmented sentence as `normal_text`, and the segmented Easy Read version as `easy_text`. Next, `rewrite_easy_segments.py` used an LLM-based rewriting pipeline to convert each Easy Read segment into a complete standalone sentence, stored under `easy_read_rewritten`. This was necessary because many Easy Read segments are grammatical fragments when read independently. Finally, `post-processing.py` flagged malformed rewritten sentences and wrote them to a separate file for manual correction or removal. After review, the corrected sentences were written back into the dataset. The final cleaned dataset, `post_processed_segmentation_all.jsonl`, contains paired standard text, original Easy Read segments, and rewritten standalone Easy Read sentences, in the following format:

```
{
  "id": int,
  "normal_text": string,
  "easy_text": list[string],
  "easy_read_rewritten": list[string]
}
```

4 Methodology

The system combines AMR graph construction, a GNN classifier trained to distinguish Easy Read from standard text, and a graph-guided LLM prompting pipeline. Each component is described as follows.

4.1 System Overview

There are two phases in our system. In the training phase, document level AMR graphs are constructed from paired standard and Easy Read texts and used to train a GNN binary classifier. In the inference phase, the trained GNN’s learned structural representations are translated into human readable guidance, which is injected into an LLM prompt alongside a ranked concept list derived from a spaCy knowledge graph. A baseline prompt comparing the same Easy Read formatting rules but without any graph-based guidance runs in parallel for comparison of output quality.

4.2 AMR Graph Construction

Each document is parsed into sentence level AMR graphs using the `amrlib` library with the pre-trained `model_parse_xfm_bart_large` model, which is an already tuned BART-large transformer. These individual sentence AMR graphs are then merged into a single document level directed graph using three types of edges: (1) semantic, connecting concept nodes according to AMR role labels (e.g., `:ARG0`, `:ARG1`); (2) discourse, connecting root nodes of consecutive sentences to preserve sequential order; and (3) coreference, connecting nodes across sentences that share the same base concept (obtained by stripping sense numbers). Node identifiers follow the convention `sentence_index::AMR_variable` to avoid cross-sentence naming collisions.

4.3 Node Feature Embedding

AMR concept strings are embedded into vectors using the `all-MiniLM-L6-v2` Sentence Transformer, producing 384 dimensional node features. Sense numbers are stripped before embedding so semantically equivalent concepts across senses receive similar representations. Each document graph is converted into a PyTorch Geometric Data object.

4.4 GNN Architecture and Training

We train a Graph Convolutional Network (GCN) to perform binary document level classification, using label 0 for standard text and label 1 for Easy Read text. The model `EasyReadGNN` consists of two `GCNConv` layers with ReLU activations, a global mean pooling operation to produce fixed size

graph level embeddings, and two fully connected layers to produce a single logit output. Dropout is applied after the second convolution and after the first fully connected layer. Training uses the Adam optimizer with a learning rate of 5×10^{-4} and weight decay of 1×10^{-4} . The loss function is binary cross-entropy with logits (BCEWithLogitsLoss). The dataset is split 80/20 into train and validation sets, and models are trained for up to 30 epochs with early stopping triggered after four consecutive epochs without improvement in validation loss, with the best checkpoint returned.

4.5 Structural Guidance

Rather than passing the raw GNN embeddings to the LLM, we translate each document graph into human readable structural statistics, including concept count, sentence count, average concepts per sentence, graph density, longest shortest path across the largest connected component, coreference edge count, and top-five highest-degree concepts. We convert these into plain English simplification instructions bench-marked against Easy Read structural baseline. For example, if the input document’s average concept count per sentence exceeds the baseline by more than 30%, the guidance explicitly instructs the LLM to split long sentences. Abstract AMR placeholder concepts (e.g., person, have, and) are also filtered from concept vocabulary passed to the LLM.

4.6 Knowledge Graph and Concept Ranking

A spaCy knowledge graph (en_core_web_trf) is constructed per document, treating named entities as single nodes and remaining content words as individual nodes. Directed edges connect each token to its syntactic head; degree centrality determines concept ranking. The top ten concepts are divided into "must-use" (above 50% of highest centrality score) and "supporting" (below 50%) tiers, both included in the LLM prompt as concept inventories.

4.7 LLM Prompt Construction and Generation

The graph-guided prompt combines (1) fixed Easy Read formatting rules, (2) structural guidance from the GNN, and (3) the ranked concept list. We use GPT-4o-mini at temperature 0.2. The baseline prompt contains only the formatting rules, applied to the same input without graph-based guidance.

4.8 Evaluation Metrics

The outputs from this system are evaluated against the Easy Read reference texts using the following metrics:

1. System output Against References and against Input (SARI) is our primary metric. It scores simplification outputs by rewarding outputs that correctly add words, keep words, and delete words, calculated as an average of n-gram F1-scores. We use the `easse` library’s `corpus_sari` implementation.
2. Bilingual Evaluation Understudy (BLEU) measures the precision of n-gram overlaps between output and reference texts, producing a score from 0 to 1.
3. Flesch-Kincaid Grade Level estimates the reading difficulty through the following formula: $(0.39 \times (\text{average sentence length}) + 11.8 \times (\text{average syllables per word}) - 15.59)$. We report this score for the original text, baseline output, graph-guided output, and the Easy Read reference.
4. Concept Preservation measures the proportion of the top five ranked concepts from the knowledge graph that appear in the system output, providing a proxy for semantic faithfulness.

Average sentence length is additionally reported to contextualize output complexity. All metrics are computed per record (sentence level) and then aggregated to corpus-level means.

5 Results

We evaluate results from the baseline prompting and our graph-guided prompting across 1909 records, where each record contains original text, an Easy Read version of the text, the original text’s AMR

graph, and the Easy Read reference’s AMR graph. Corpus-level means for all of the metrics are reported in Table 1, with standard deviations reported in Table 2. We organize findings by metric below, presenting corpus-level aggregates alongside their corresponding per-record distributions.

Table 1: Corpus-level mean metrics, including SARI, BLEU, Flesch-Kincaid grade, concept preservation, and average sentence length.

Metric	Baseline	Graph-guided	Δ
SARI	10.156	15.009	+4.853
BLEU	0.143	0.255	+0.112
FK Grade (output)	6.745	6.839	+0.094
FK Grade (original)	10.662	–	–
FK Grade (reference)	–	10.662	–
Concept Preservation	0.637	0.888	+0.251
Avg. Sentence Length	6.271	5.898	-0.372

Table 2: Corpus-level mean and standard deviation for evaluation metrics across baseline and graph-guided outputs.

Metric	Baseline	Graph-guided
SARI	10.156 $\pm$ 5.162	15.009 $\pm$ 5.451
BLEU	0.143 $\pm$ 0.143	0.255 $\pm$ 0.177
FK Grade (output)	6.745 $\pm$ 4.158	6.839 $\pm$ 3.999
Concept Preservation	0.637 $\pm$ 0.267	0.888 $\pm$ 0.150
Avg. Sentence Length	6.271 $\pm$ 1.581	5.898 $\pm$ 1.502

5.1 SARI

The graph-guided system achieves a mean SARI of 15.009, compared to 10.156 for the baseline. This improvement is statistically significant by a paired t-test ($t = 40.687$, $p < 0.001$). Per-record SARI scores are visualized in Figure 1, which shows that the graph-guided output was consistently above the baseline across the corpus. The per-record win/loss/tie breakdown in Table 3 confirms that the graph-guided output outperforms the baseline on 1547 records, while the baseline only outperforms the graph-guided outputs on 173 records.

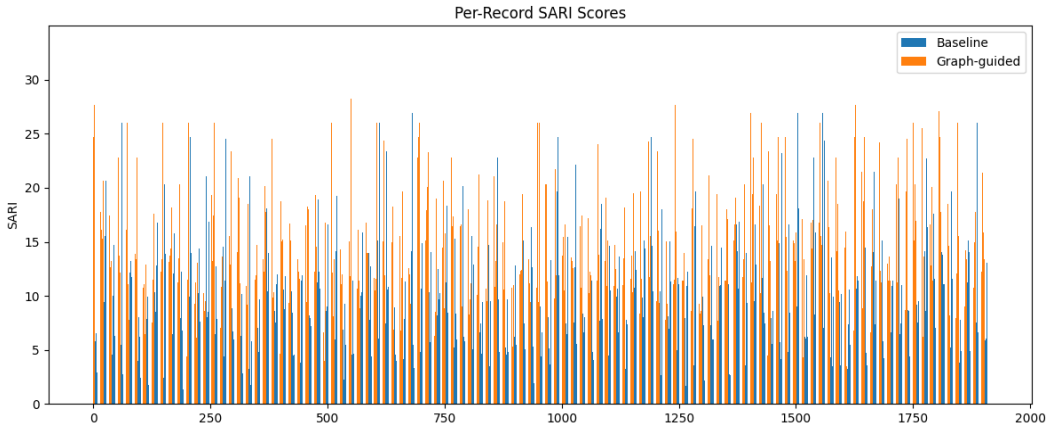


Figure 1: Per-record SARI scores for baseline and graph-guided outputs.

Table 3: Per-record SARI win, loss, and tie counts comparing baseline and graph-guided outputs.

Outcome	Count	Percentage
Graph-guided wins	1547	81.0%
Baseline wins	173	9.1%
Ties	189	9.9%

The SARI improvement demonstrated in Figure 2 plots the per-record delta (graph-guided SARI minus the baseline SARI), and the mean delta is $+4.853$ with a median of 4.39 , which shows the improvement is distributed across the records broadly.

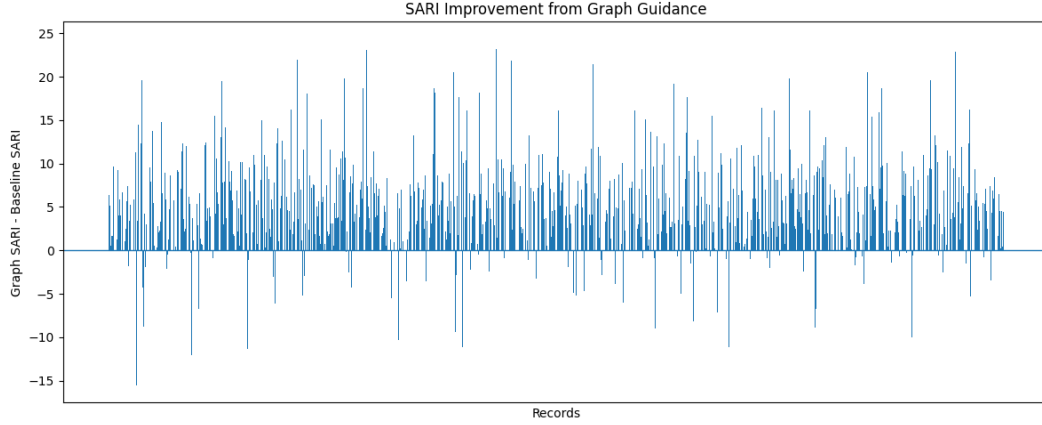


Figure 2: SARI Improvement from graph guidance.

5.2 BLEU

Figure 3 shows the distribution of per-record BLEU scores for both conditions. The baseline distribution is concentrated heavily near zero with a mean of 0.143 , whereas the graph-guided distribution has considerably more records in the $0.2 - 0.5$ score range and a mean of 0.255 . The graph-guided outputs not only raises the average and reduces portion of near zero records, which suggests its outputs are more consistently similar to the Easy Read references. The findings are statistically significant ($t = 25.561$, $p < 0.001$).

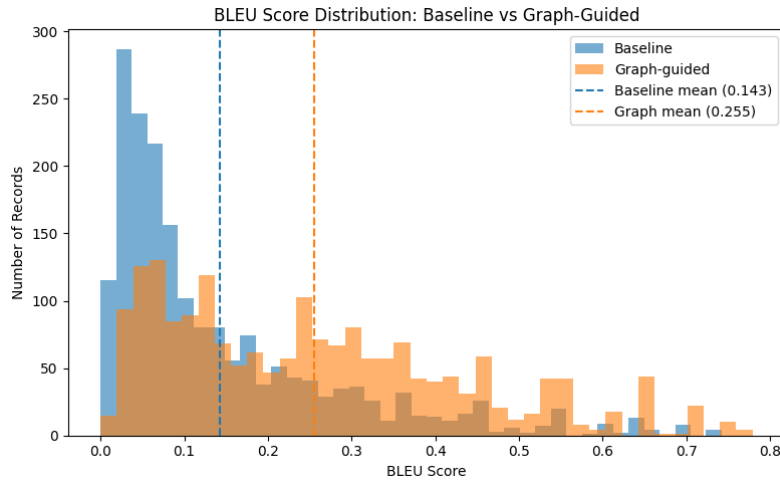


Figure 3: BLEU Distribution for baseline and graph-guided outputs.

5.3 Readability

Both baseline and graph-guided outputs reduce the reading difficulty relative to the original texts, as shown in the Flesch-Kincaid readability comparison in Figure 4.

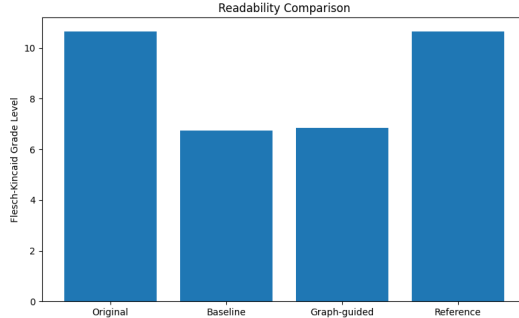


Figure 4: Flesch-Kincaid Scores for the original record text, reference Easy Read text, baseline responses, and graph-guided responses.

The baseline output achieves a mean grade level of 6.745 and the graph-guided output achieves a 6.839, compared to the original mean of 10.663. The 0.094 difference between conditions is not statistically significant ($t = 1.365$, $p = 0.1723$), indicating that the graph-guided system neither helps or hurts readability relative to the baseline.

5.4 Concept Preservation

The graph-guided system preserves a significantly higher portion of top ranked concepts. Mean concept preservation improved from 0.637 at baseline to 0.888 in graph-guided outputs, which is statistically significant ($t = 43.338$, $p < 0.001$). This information is expanded upon in Table 4.

Table 4: Concept preservation results comparing baseline and graph-guided outputs.

Metric	Baseline	Graph-guided
Mean $\pm$ SD	0.637 ± 0.267	0.888 ± 0.150
Records with perfect preservation (≥ 1.0)	21.3%	57.0%
Records above baseline	–	64.2%
Records below baseline	–	0.7%

6 Discussion

The results demonstrate that graph-guided prompting substantially improves Easy Read text simplification quality over a rules-only baseline. The improvements in SARI (+4.853) and BLEU (+0.112) are both statistically significant and distributed across the corpus, with the graph-guided outputs outperforming the baseline on 81% of records. This suggests that translating AMR graph statistics into human-readable simplification guidance provides the LLM with meaningful constraints that better align the outputs with Easy Read conventions. The largest gains appear in concept preservation (+0.251), which is expected since the ranked concept list from the spaCy knowledge graph is directly referenced in the prompt.

However, the readability results show less promising results. While both graph-guided and baseline outputs reduce Flesch-Kincaid grade level relative to the original texts, neither approaches the reference Easy Read grade level, and the difference between the FK scores is not significant. This might reflect a limitation on the use of GPT-4o-mini, which might produce shorter sentences without reducing complexity of vocabulary. Alternatively, an additional step with LLM embeddings for semantic simplification could also help reduce the output sentences’ grade level.

We note that SARI scores for both graph-guided outputs and baseline outputs remain low. Since Easy Read texts can be simplified in many ways, human evaluation and multiple references should be used in the future to assess the quality of the output, rather than relying on automatic metrics. The lack of human verification is a major limitation.

There are several other limitations that reduce the generalization of these findings. It is not clear how well the AMR parsing and GNN classifier might transfer to medical, legal, or other domains where Easy Read is common. Additionally, our reliance on GPT-4o-mini limits reproducibility and imposes costs that might not be replicable at scale. Future work could extend the GNN classifier to provide finer-grained guidance beyond binary Easy text or standard text classification. Future work could also explore the generation of images in Easy Read texts. Images are common in Easy Read texts for easier understanding, and our work does not touch on image generation or sourcing at all.

7 Broader Impact

This project aims to improve access to written information for people with intellectual disabilities and other readers who may benefit from Easy Read text. Although Easy Read has important implications for information accessibility, relatively little research has examined how recent advances in machine learning, graph-based representation learning, and large language models can support its automation. This project aims to address this gap in research. If developed responsibly, graph-guided simplification systems could help organizations create more accessible versions of public, educational, legal, medical, and civic documents at lower cost and greater scale. However, automated Easy Read generation also carries risks. Poor simplification can remove important information, introduce misleading phrasing, or produce text that appears accessible while failing to meet the needs of its intended audience. Because Easy Read is normally validated by trained specialists and members of the target community, systems like ours should be treated as assistive tools rather than replacements for human review. Future work should include evaluation with Easy Read experts and readers with intellectual disabilities, as well as attention to layout, images, and other accessibility requirements beyond textual simplification.

8 Conclusion

This project shows that AMR-derived graph structure can improve LLM-based Easy Read simplification. By training a GNN over document-level AMR graphs and converting graph-derived features into prompt guidance, our system improves SARI, BLEU, and concept preservation over a baseline LLM prompt while maintaining comparable readability. These results suggest that graph-guided prompting is a promising direction for more faithful and interpretable Easy Read automation. Future work should evaluate outputs with Easy Read experts and target readers, and extend the system beyond textual simplification to layout, images, and human validation.

9 Contributions

Ava Baraban contributed to data preprocessing and the literature review. Nomin Rentsendorj contributed to GNN model development and benchmarking. Both authors contributed to the design of the proposed graph-guided simplification pipeline and to AMR generation.

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